CS-E407523 – Machine Learning Applications in Universe Sciences (MLUS)

A1: **Update regarding submissions**

Dale Weigt

Maarit Korpi-Lagg

dale.weigt@aalto.fi

Aalto University
School of Science

MyPreprocessor class

- **Initially: class was defined in your cell, saved in .joblib bundle.**

MyPreprocessor class

- **Initially: class was defined in your cell, saved in .joblib bundle.**
- ******SLIGHT CHANGES HERE******

MyPreprocessor class

- **Initially: class was defined in your cell, saved in .joblib bundle.**
- ***SLIGHT CHANGES HERE***
  - Still utilising your class objects but in a different way
  - Need them to be saved in a **separate .py file in your folder named "preprocessor_class.py"**
  - This class is then called from **the top of the cell you save your model**

Example

/ ··· / mlus2026 / Assignment_1 /

Name	Last Modified
A1_Auroral_boundaries.ipynb	an hour ago
earth_msphere.png	7 days ago
logo_v1.png	7 days ago
preprocessor_class.py	seconds ago
student_model.joblib	an hour ago

- **Folder should contain these files** before submission
- **Additional files** (i.e. showing additional work, feature engineering…) also fine
- ***If your submission does not contain these files on the left, it will not be graded for the scoreboard***

Example

Filter files by name

/ ··· / mlus2026 / Assignment_1 /

Name	Last Modified
A1_Auroral_boundaries.ipynb	an hour ago
earth_msphere.png	7 days ago
logo_v1.png	7 days ago
preprocessor_class.py	a minute ago
student_model.joblib	an hour ago

```python
import pandas as pd


class MyPreprocessor:
    def __init__(self, feature_columns):
        self.feature_columns = list(feature_columns)

    def transform(self, df):
        df = df.copy()
        X = df[self.feature_columns]

        # Drop rows with missing values
        keep = X.dropna().index
        self.keep_index_ = keep

        return X.loc[keep]
```

- **Example of preprocessor_class.py file**

Example

```
[ ]: import joblib
     from preprocessor_class import MyPreprocessor
     print(MyPreprocessor.__module__)  # should be preprocessor_class
```

- **Make sure you import your class before saving your model**

```
     """ADD MODEL AND OTHER FEATURE ENGINEERING ETC. HERE"""
```

- **REMOVE class object from your cell here**

```
     # ----------------------------
     # SAVE SUBMISSION BUNDLE
     # ----------------------------
     bundle = {
         "target": TARGET,                    # student declares target
         "model": clf,                        # trained estimator
         "preprocessor": MyPreprocessor(FEATURES),
     }
```

- **Save model the same way as before**

```
     joblib.dump(bundle, "student_model.joblib")
     print("Saved student_model.joblib")
```